

can't rebalance, hold the market. (The horserace results in Table 3.14 show that you will do pretty well and much better than mean-variance.)

If you can rebalance, move to the equal weight portfolio. You will do better than market weights in the long run. Equal weights may be hard to implement for very large investors because when trades are very large, investors move prices and incur substantial transaction costs. It turns out that any well-balanced, fixed-weight allocation works well. Jacobs, Müller, and Weber (2010) analyze more than 5,000 different portfolio construction methods and find that any simple fixed-weight allocation thrashes mean-variance portfolios.

Now if you can estimate variances or volatilities, you could think about risk parity. My horserace only estimated volatility by taking the realized volatility over a past rolling sample. Ideally we want estimates of future volatility. There are very good models for forecasting volatility based on generalized autoregressive conditional heteroskedasticity or stochastic volatility models, which I describe in chapter 8.

Suppose the hotshot econometrician you've just hired can also accurately estimate correlations and volatilities. Now you should consider relaxing the correlation restriction from risk parity. Finally, and hardest of all, is the case if you can accurately forecast means. If and only if you can do this, should you consider doing (fairly unconstrained) mean-variance optimization.

Common to all these portfolio strategies is the fact that they are diversified. This is the message you should take from this chapter. Diversification works. Computing optimal portfolios using full mean-variance techniques is treacherous, but simple diversification strategies do very well.

Warning on Risk Parity

The second panel of Table 3.14 reports the average weights in each asset class from the different strategies. Risk parity did very well, especially the risk parity strategy implemented with volatilities, because it overweighted bonds during the sample. Risk parity using variances held, on average, 51% in U.S. Treasuries and 36% in corporate bonds versus average market weights of 14% and 8%, respectively. There were even larger weights on bonds when risk parity is implemented weighting by volatilities rather than variances. Interest rates trended downward from the early 1980s all the way to the 2011, and bonds performed magnificently over this period (see chapter 9). This accounts for a large amount of the out-performance of risk parity over the sample.

Risk parity requires estimates of volatilities. Volatilities are statements of risk. Risk and prices, which embed future expected returns, are linked in equilibrium (see chapter 6). Howard Marks (2011), a hedge fund manager, says: "The value investor thinks of high risk and low prospective return as nothing but two sides of the same coin, both stemming primarily from high prices." Risk parity overweights assets that have low volatilities. Past volatilities tend to be low precisely